Transferable Skills Matching: Explainable, Human-in-the-Loop Proposal

2026-10-04 · @u_0TUetmjeUqSAhE6317KWhA

1. Summary

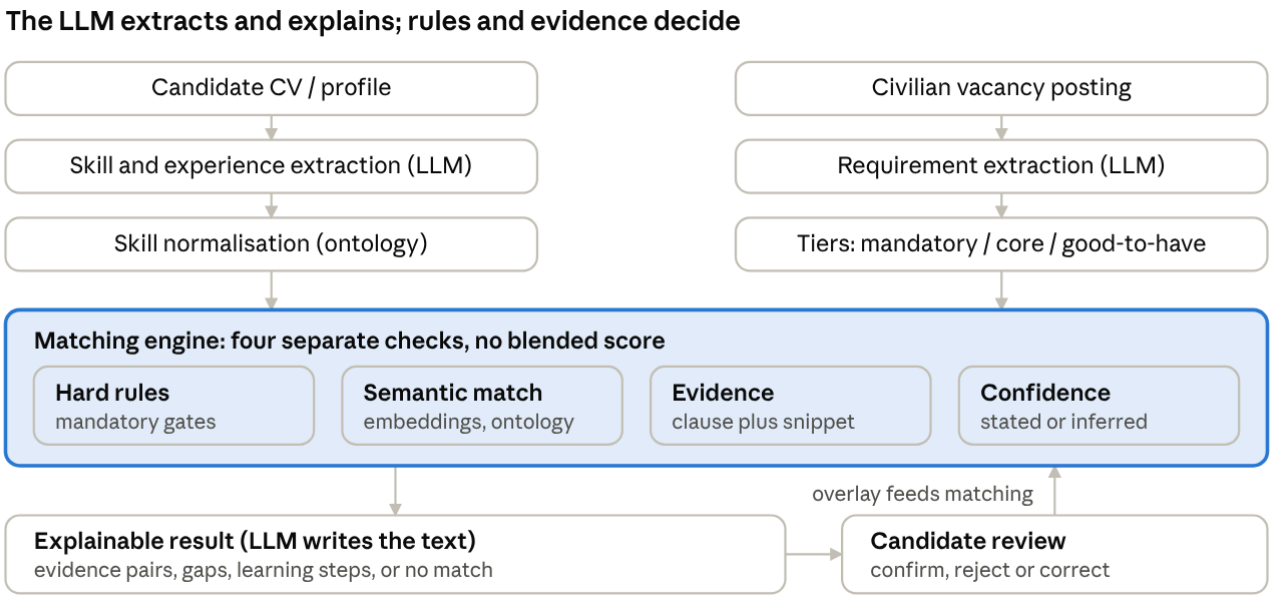
Problem. Military personnel hold real skills that civilian vacancies describe in other words, so keyword matchers miss genuine fits and force false ones. The gap is legibility, not capability: in a US study, about 86% of military occupations overlapped moderately to strongly with civilian ones, yet about 41% of post-9/11 veterans perceived themselves as underemployed (Journal of Veterans Studies; US data, no EU equivalent found).

Main idea. An explainable, human-in-the-loop tool that translates military experience into civilian opportunities. It recommends and explains, and never decides hiring or eligibility. Each match pairs a vacancy clause with candidate evidence, mandatory requirements are checked by rules apart from skills, "no suitable match" is a valid answer, and candidates see their gaps before proceeding and can correct the record.

Intended users. Service leavers; NSPA-facilitated advisers, who review; employers and vacancy authors (opt-in only).

Expected benefits (assumptions). Faster discovery of relevant roles, fewer applications to roles with unmet mandatory requirements, a clear path for each gap, and vacancy wording audited for hidden filters.

2. AI Approach



The LLM extracts and explains; rules and evidence decide.

1. **Extract.** An LLM (Claude or Gemini API) turns the CV into a structured schema (roles, tools, certifications, languages, years worked per skill area), each item tied to its source snippet. Medical, religious, union and ethnicity content is redacted first. Matching scores the schema, not the prose.
2. **Normalise.** Skills map to ESCO; the ESCO to O*NET crosswalk reaches the* O*NET military crosswalk; embeddings match equivalent wording.
3. **Ingest vacancies.** Requirements are tiered mandatory, core or good-to-have. Each eligibility category is declared with a learnable flag and a gate type (negotiable, fixed, unclear). Proxy wording such as "native speaker" or an age cap is flagged to the vacancy author, not applied.
4. **Match.** Rules check mandatory gates (met, not met, unverified). Semantic matching checks core skills (stated, inferred, not found). Every claim carries an evidence pair. There is no blended score.
5. **Explain.** The LLM writes the explanation from these results and never decides.

Result state	Rule
Potential match	No mandatory requirement unmet; at least two-thirds of core skills evidenced
Potential match, mandatory gap	Core at least two-thirds; a learnable mandatory requirement unmet, shown first
Skills match, blocked by fixed requirement	Core at least two-thirds; a non-learnable gate unmet, quoted, no advice given
No suitable match	Core below two-thirds

Inferred skills count only once the candidate confirms them, and unverified never counts as failed. Two-thirds is a starting value, tuned in the pilot.

Data sources. ESCO v1.2.1 (13,939 skills, 3,039 occupations, 28 languages) and O*NET 31.0, both free with attribution; fictional profiles and vacancies in the prototype; volunteers' CVs and an NSPA-supplied vacancy set in the pilot.

People. Candidates confirm or reject inferred skills and acknowledge gaps. Trained NSPA advisers review a sample and can override any recommendation. Employers make every hiring decision.

3. Fairness

Reducing bias.

- Gender, age, ethnicity, religion, marital status, photo and health data never enter matching. Gates such as citizenship are asked only when a vacancy declares them, are checked as separate rules quoting the vacancy clause, and never feed skill scoring.
- Removing names is not enough: in one study, LLM scores differed by readability and sentence length ($p < 0.001$) while name substitution had no significant effect (Rao et al.). The system scores the structured schema, and tests an identity swap and a same-facts rewrite in plainer or non-native phrasing. Result states must not change under either.
- Career gaps were the second-largest penalty in a 25,500-evaluation LLM study (re-cinq, not peer reviewed). The system records years worked, not dates, which raised callbacks by 15% for CVs with gaps in a field experiment (Kristal et al.), and keeps military training separate from certification.
- Published name-bias findings conflict, so bias is measured on our own model and prompts. Optional demographic data, given with explicit consent and held outside the matching pipeline, is used only to monitor the pilot.

Understandable outputs. A readiness view, not a score: for example 4 mandatory requirements met, 1 not met and 1 unverified, then core skills evidenced, awaiting confirmation or not found. Each line expands to its vacancy clause and candidate snippet, and "not found" never means "does not have". Plain language, targeting CEFR B1 (ISO 24495-1).

Accessibility. A voluntary commitment to WCAG 2.2 AA; the binding reference remains EN 301 549 v3.2.1, WCAG 2.1 AA (source). Status is shown by text and icon, not colour alone; the results screen is tested with keyboard and screen reader, on mobile and at low bandwidth; multilingual output builds on ESCO's 28 languages; an external audit precedes the pilot.

Error correction. Candidates add experience, challenge inferred skills, and confirm or reject suggested transfers. Corrections form a timestamped overlay that never alters the source CV. Under warn and proceed, candidates see every mandatory gap before applying, can fix items and re-run the check, then acknowledge what remains, and the acknowledgement is logged. Fixed gaps are stated plainly, with no advice that could not help.

Edge cases. Restricted (classified) experience is marked self-attested, never counted as met and never as absent. Career breaks and military courses without a civilian certificate are not penalised. A strong candidate blocked only by a negotiable gate, such as location, is visible to an employer only on separate opt-in, showing the skills evidence and the named gate, never nationality.

4. Plan

A three-month pilot, facilitated by NSPA and built and run by the entrant alone. Assumed scope: 20 to 30 volunteer service leavers and 30 to 50 civilian vacancies, with NSPA advisers reviewing.

Weeks	Work	Responsible
1-2	Agree scope, data terms and the vacancy source; start the combined DPIA and impact assessment	Entrant; NSPA sponsor and data-protection contact
3-6	Build and harden; prototype with three fictional profiles (two matches, one correctly returning "No suitable match identified")	Entrant
5-6	Checks before deployment, below	Entrant; external reviewers; counsel
7-12	Run the pilot with weekly log review and adviser review of sampled recommendations; evaluate against section 5 in week 12	Entrant; NSPA advisers

Checks before deployment (any failure blocks go-live): DPIA and impact assessment reviewed by counsel and NSPA's data-protection contact; external accessibility audit with no blocking findings; unchanged result states under both fairness perturbations; no unsupported claim in sampled explanations; security review with prompt-injection tests; NSPA-approved candidate notice and consent flow.

Resources. A small EU-region cloud app with Postgres and pgvector, the Claude or Gemini API, the ESCO and O*NET data, and adviser and participant time from NSPA (assumption).

Cost line (3 months)	Estimate	Basis
Technical: LLM inference, embeddings, hosting, pgvector, ESCO and O*NET	EUR 50-500	Published prices: inference $3-30 at about 500 explanations a month, embeddings under $1, hosting EUR 15-180, data free; rounded up for headroom
External accessibility audit	EUR 1,400-3,700	Published UK G-Cloud rate of GBP 400-800 a day for 3-4 days, at about EUR 1.17 per GBP
External fairness review	EUR 1,400-4,700	Same day rates for 3-5 days; inferred, as no published bias-audit rate was found
Legal and data-protection review	EUR 1,500-4,000	Placeholder, to be quoted
Security review	EUR 500-2,000	Placeholder, to be quoted
Developer, adviser and participant time	In kind	Developer valued at the published $55-110 an hour benchmark; hours to be set

Cost line (3 months)	Estimate	Basis
Cash total	**EUR 4,900-14,900**	Sum of the cash lines before a 15% contingency (assumption); prices checked 4 October 2026

5. Impact

Three indicators are measured in the pilot. Every target is an assumption, and the pilot also sets the baseline.

Indicator	How it is assessed	Target (assumption)
1. Relevance and gate accuracy	Advisers rate a blind sample of recommended roles, and compare their reading of each sampled vacancy's mandatory requirements with the system's	80% of roles rated relevant; 95% of mandatory requirements detected
2. Time to shortlist	Median minutes to reach three roles worth applying to, tool versus unaided search, same cohort	50% shorter
3. Fairness stability	Share of test profiles with unchanged result states under both perturbations; correction rates monitored for patterns	100% unchanged; any change fixed first

With 20 to 30 participants, results are indicative. Unsupported-claim and false-positive rates are tracked as safety counters, and the benefits in section 1 stay assumptions until measured.

6. Risks

This is not legal advice; counsel review is budgeted in section 4.

Risk	What we do
AI Act. The system is high-risk under Annex III point 4(a) (analysing applications, evaluating candidates). The Article 6(3) exemption fails because ranking is evaluative and profiling is excluded from it. Human oversight meets Article 14 but does not change the class	Regulation (EU) 2026/1744, the AI half of the Digital Omnibus, defers Annex III duties to 2 December 2027. We build to that baseline now: self-assessed conformity (Article 43, Annex VI), logging, oversight and documentation. No harmonised standard is yet cited
GDPR Article 22. Employer-facing output could count as an automated decision, since a score a human relies on heavily can itself be one (*SCHUFA*, C-634/21). The GDPR half of the Omnibus (2025/0360(COD)) is still only a proposal, so GDPR applies unamended	Unranked, evidence-annotated sets with no score or auto-rejection, under explicit consent per employer. The evidence pair is designed to meet the explanation standard in *Dun & Bradstreet* (C-203/22), as a design commitment and not certified compliance
Privacy. Military CVs can indirectly reveal health, religion, union membership or ethnicity (*OT*, C-184/20), and inference can create such data	Redaction before the model; no inference of special-category or fitness traits; raw service and discharge records never ingested; eligibility data asked per vacancy; tiered retention; separate employer-sharing consent; combined DPIA and impact assessment before go-live
Security. Sensitive data, and prompt injection in CV or vacancy text	Encryption, role-based access, audit logs, EU-region hosting; CV and vacancy text treated as untrusted; the LLM never decides eligibility; unsupported claims are dropped
Ethics. Wrong output affects real careers	No hiring or eligibility decisions and no advice to waive legal requirements; warn and proceed is logged; bias measured per model; false positives reported
Challenge-specific. No public EURES vacancy API was found (EURES). NSPA's status under GDPR and AI Act Article 27 is open, and if NSPA offers the tool to its own departing personnel, the consent analysis reverts to the employment case	Fall back to an NSPA-supplied or curated vacancy set (the largest dependency); assume the stricter standard and seek counsel; keep the operator apart from chains of command; never condition access on employer sharing

7. Tools

Tools used. Claude for deep research, drafting, the architecture diagram and the prototype; ChatGPT for supplementary document research; the Claude or Gemini API for prototype inference (choice open; Gemini's Flash-tier price doubles on 1 January 2027, immaterial at pilot volume); an embedding model still to be selected, costed at the published text-embedding-3-small rate; a small cloud app with Postgres and pgvector; ESCO v1.2.1 and O*NET 31.0.

Fictional data, assumptions and mock-ups. All profiles (A, B and C), vacancies and the employer view are fictional, and no real personal data is used. The prototype is a mock-up of the method, not a production system. Pilot size, targets, adviser time and the legal and security costs are assumptions or placeholders, and all benefits are projections. AI-assisted research was checked against the sources below, and unverified points are marked open in the text.

Sources. Regulation (EU) 2026/1744; AI Act Annex III, Article 6, Article 43; procedure 2025/0360(COD); CJEU rulings via secondary summaries (SCHUFA, Dun & Bradstreet, OT); Rao et al.; re-cinq; Kristal et al.; Journal of Veterans Studies; EN 301 549 update; ESCO; O*NET. Prices checked 4 October 2026: Anthropic, Gemini, OpenAI embeddings, Railway, Render, G-Cloud.